

Feasibility and Cost Optimization Study of Landfill Gas to Energy Projects Based on a Western Cape Landfill Site in South Africa

W.Nguz Mbav
University of Cape Town,
South Africa
jwilfriednguz@hotmail.fr

S.Chowdhury
University of Cape Town,
South Africa
sunetra.chowdhury@uct.ac.za

S.P.Chowdhury
University of Cape Town,
South Africa
sp.chowdhury@uct.ac.za

Abstract – One of the serious problems that the world still faces nowadays is the chronic power shortage and the inaccessibility to energy services of mostly the majority of the population living in remote or off-grid areas. In addition to this concern, rapidly growing populations, rapid economic growth and rise in community living standards have accelerated the generation rate of Municipal Solid Waste (MSW) causing its management to be a major worldwide challenge. This paper deals with the technical, economic, and environmental assessment of deploying a Landfill Gas to Energy generation system considering the Bellville South Landfill Site (South Africa) as a case-study. The research work also aims to introduce systems that enable to generate energy; analyze quantity and economic value of decomposition products (gas); quantify estimated capital and operating costs by performing an economic assessment of an LFGTE Projects using microturbine technology rather than the mostly and commonly worldwide used IC Engines technology. The work also brings to light the relevant environmental impacts. Furthermore, the economic assessment in this research work is performed using the Hybrid Optimization Model for Electric Renewables (HOMER) software in order to develop the cost optimization analysis.

Index Terms– decomposition products, economic and environmental impacts, electricity generation, landfill gas, solid wastes

I. INTRODUCTION

The fact that the energy sector is closely interwoven with the economy highlights the ultimate evidence of a two-way linkage or relationship between energy and economic development. This also shows and reaffirms the pivotal role of energy, particularly the desired electricity in improving living standards and sustaining economic growth.

The combination of burgeoning manufacturing sector, rapid population growth, rapid urbanization and spiraling domestic consumption has led nations around the world to facing the difficult challenge of generation, treatment, and disposal of solid wastes. This has also left nations with the option to explore the full spectrum of potential alternative energy solutions. So, in a bid to promote sustainable development and at the same time, diversify the nation's energy matrix, the concept of integrated waste management has evolved from an intensive waste recycling program to energy production from the Landfill Gas (LFG). And this has changed the perception of Landfill Gas seen as a liability to be an asset.

As at present, cost is a major factor in determining the choice of any alternative energy solutions, an economic analysis using HOMER software will be performed. This paper deals with the technical, economic and environmental assessment of deploying a Landfill Gas to Energy (LFGTE) generation system considering the Bellville South Landfill Site as a showcase.

II. BELLVILLE SOUTH LANDFILL SITE DESCRIPTION

The Bellville landfill is situated in City of Cape Town municipality and was established in 1960. The site is located at 33°56'07.0" S, 18°38'57.2" E, approximately 25 Km East of Cape Town city centre. It is classified as a GLB+ (General Waste, positive climatic water balance large site) landfill, according to the license issued by DWAF [1]. The Bellville landfill has no history of odor complaints, dust complaints or fires. The geology at the site comprises a white sandy soil typical of the Cape flats, estimated to be around 30m in thickness. Layers of concrete are also typical of the Cape Flats sand deposits (leading to perching of groundwater where present). The sands are likely to be underlain by Malmesbury Shale.

III. WASTE GENERATION AND COMPOSITION

A. Waste generation

Besides physical and chemical parameters, landfill gas generation model will require annual incoming waste tonnages (for every year until projected closure) as input data. It is noted that the City of Cape Town has estimated the operational start date for Bellville South landfill site as the "early 1960's" (consequently assumed by WSP to be January 1960) whereas the available weighbridge data only goes as far back as 1999.

In addition, the City of Cape Town has provided the results of a site survey of the Bellville landfill undertaken in mid-2009. The survey has estimated a monthly usage (of available airspace) of 46,263 m³, equivalent to 41,637 tonnes at 0.9t/m³. In absence of additional and more reliable data, WSP has adopted this value (multiplied by 12) for 2009 and assumed an annual growth rate of 3% per annum for projecting historical and future waste tonnages [7]. Therefore, the following assumptions have been made when estimating waste tonnages:

- Start date: January 1960.
- End date: December 2015 (based on available airspace estimate from survey) a reference tonnage of 499,644 tonnes for 2009, a growth rate of 3% per annum, and a density of 0.9t/m³.
- From the above, the total volume of waste is estimated at 18.4 million m³, while the site is reported to have a maximum capacity of around 20 million m³. The remaining 12% or so is reasonably assumed to be required for capping, cover material etc.

Based upon the above mentioned assumptions, the following equation [14] is used to derive historical and future waste tonnages coming into the site:

$$t_n = t_i(1 + r)^n \quad (1)$$

where:

- t_n : is the waste tonnage of the nth year;
- t_i : is the initial waste tonnage;
- r : is the annual waste growth rate;
- n : is the number of years of waste placement.

B. Waste Composition of Bellville South Landfill Site

As there is insufficient data available to determine a site specific waste composition for the Bellville South, waste composition data for Coastal Park and Vissershok landfill sites have been used in this work. The simple average calculated based on the two waste streams is used to determine the Bellville South waste composition. This is suggested because all of the three landfill sites present the same features and are bound by the same climatic conditions. Based on the information contained in the waste characterization study of the Coastal Park and Vissershok landfill sites, the waste composition presently at the Bellville South landfill site is present in the table below as follows:

TABLE I
WASTE COMPOSITION FOR BELLVILLE SOUTH BASED ON THE COASTAL PARK AND VISSERSHOK 2007 SURVEY

Category	% Total Annual waste-Bellville South	% Total Annual Waste-Vissershok	% Total Annual Waste-Coastal Park ¹
Paper	19%	15.6%	22.4%
Glass	7.6%	6.2%	8.9%
Metal	4.3%	3.3%	5.3%
Plastic	9.8%	7.6%	11.9%
Textile	4.0%	5.5%	2.4%
Organic – Food	4.1%	6.5%	1.8%
Organic – Garden	5.9%	6.3%	5.4%
Wood Waste	0.6%	0.6%	-
Rubber, leather, bones etc.	0.5%	0.5%	-
Other organics	8.6%	8.6% ³	-
Household ⁴	14.9%	13.6%	16.2%
Rubble	12.7%	-	12.7%
Mixed ³	9.7%	10.0%	9.4%
Other Inorganic	-	-	-
Other	9.7%	15.8%	3.6%

¹ City of Cape Town Baseline Landfill Gas Study Emissions Curve: Coastal Park. WSP, 2009 (WSP ref. 90223CL)

² Source: Mexico Landfill Gas Model v.2, US EPA 2009

³ Waste comprises sludge, oil interceptor liquids, oil and solvent residues, yeast etc.

⁴ Mixed domestic refuse brought in private vehicles

⁵ Wood and garden waste mixed with sand, or rubble mixed with sand/general waste

C. Landfill Gas Production Modeling

A landfill gas model is a tool used to project methane generation over time from a mass of waste. In its simplest form, a model predicts methane generation or recovery from a single batch of waste, landfilled at a single given point in time. Then the total methane generation is the sum of outputs from all batches in the landfill. Several methods have been described for modeling landfill gas formation [5-6]. In general, landfill gas (LFG) formation models are not based on microbiological or biochemical principles, but more on a practical description formation, as it can be observed in laboratory experiments or in full-scale recovery projects. LFG is formed as a result of biodegradation of the organic carbon in the waste: per kg of organic carbon that degrades, about 1.87 m³ of landfill gas normalized to 1 atm and 0 °C is produced [5]. Furthermore, although the theoretical quantity of landfill gas from a ton of typical domestic refuse is around 500-600 m³ at 50% methane concentration, the conditions within any given landfill site are unlikely to be ideal.

The output of a gas model provides the starting point for the development of a project. The quantity and quality of landfill gas production will vary from site to site. In fact, several methods for modeling landfill gas production are possible, such as, Empirical method for calculation of biogas production (considering time), First Order Decay model (FOD), the Triangular model, the Scholl Canyon model, etc. However, for the purpose of this study, the US EPA's Landfill Gas Emission Model (LandGEM Version 3.02) model adopted by WSP for the evaluation of landfill gas emissions is used. LandGEM is a Microsoft Excel-based emissions modeling tool that can be used to estimate emission rate for total landfill gas (LFG), methane (CH₄), carbon dioxide (CO₂), non-methane organic compounds (NMOCs), and individual air pollutants from municipal solid waste landfills.

The model is intended for large engineered landfills (as opposed to shallow unmanaged dumps) such as the Bellville South, Coastal Park and Vissershok landfills, where anaerobic biodegradation is the dominant mechanism for landfill gas production. But the basis for selecting the LandGEM model is its extensive track record internationally, specifically for air quality assessment, and its transparent methodology. LandGEM is based on a relatively simple first order decomposition rate equation for quantifying emissions from landfilled municipal solid waste. According to reference [2], the gas production rate is assumed to decrease as the organic fraction of the refuse diminishes. The key equation governing this first order decay model is described as follows:

$$Q_{ch4} = \sum_{i=1}^n \sum_{j=0.1}^1 kL_0 \left[\frac{M_i}{10} \right] e^{-Kt_{ij}} \quad (2)$$

Where:

- Q_{ch4} = annual methane generation in the year of the calculation (m³/year)
- i = l year time increment

- n = (year of the calculation) – (initial year of waste acceptance)
- $j = 0.1$ year time increment
- K = methane generation rate (year^{-1})
- L_0 = potential methane generation capacity (m^3/Mg)
- M_i = mass of waste accepted in the i -th year (Mg)
- t_{ij} = age of the j -th section of waste mass M_i accepted in the i -th year (decimal years)
- Mg = mega-grams mass unit equivalent to 1 ton or 10^6 grams

D. Landfill gas Composition and Characteristics

Landfill gas is generated through the anaerobic decomposition of organic waste present in municipal solid waste. The landfill design and operation contributes to the decomposition process. Generation starts shortly after a landfill begins receiving waste and can last for up to 30 years after the landfill closes. The average composition of this landfill gas (LFG) is about 50% methane (CH_4), 45% carbon dioxide (CO_2), and 5% nitrogen (N_2) and other gases [3]. There are also trace amounts of non-methane organic compounds (NMOC).

Based on the results of the 2007 waste characterization surveys for the City of Cape Town, the physiochemical model input parameters such as LFG methane content, methane generation capacity (L_0), and methane generation rate (K) are summarized in the table below:

TABLE II
MODEL INPUT PHYSIOCHEMICAL PARAMETERS

Methane Content [% by volume]	Methane Generation Rate, (K) [year^{-1}]	Potential Methane Generation Capacity, (L_0) [m^3/Mg]
58.3%	0.05	126

V. LANDFILL GAS AND METHANE EMISSIONS CURVE

Depending on the characteristics of the waste mass and the volume of the landfill, a sufficient volume of “green” energy in the form of methane is generally expected to be produced during operations, such that beneficial use is economical. Based on the input parameters related to the annual waste tonnages entering the landfill site, the potential methane generation capacity (L_0) and the methane generation rate (K), as described and summarized in table III, the resulting emissions evolution curve, and current and predicted peak emissions are shown and tabulated here below:

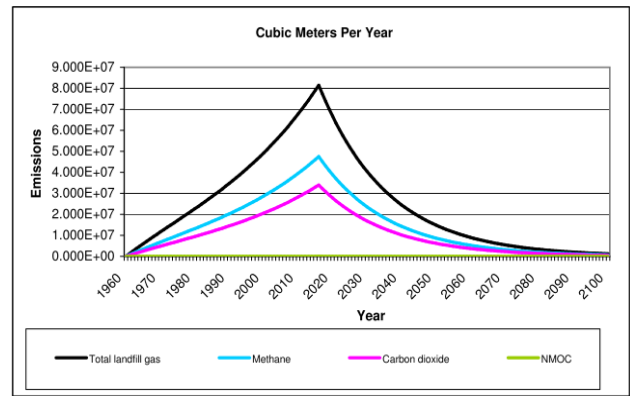


Fig.1. Landfill gas and methane emissions curve for Bellville South Landfill Site

TABLE III
CURRENT AND PREDICTED PEAK EMISSIONS FOR 2009 AND 2016
RESPECTIVELY

Year	Waste-in-place [tonnes]	Total Landfill Gas [m^3/yr]	Methane [m^3/yr]	NMOC's ¹ [m^3/yr]
2009	13,241,344	65,670,000	38,280,000	39,400
2016	16,570,190	81,480,000	47,500,000	48,890

IV. LANDFILL GAS TECHNOLOGY OPTIONS

Due to the inherent presence of contaminants and trace gases alongside with LFG, and which can limit and affect the efficiency of the technologies applied, it is suggested LFG to be pre-processed to remove the vast majority of the CO_2 and non-methane organic constituents prior to its utilization. And the main three clusters of processing that LFG undergone before its utilization are described hereafter:

- *Low-Grade LFG*: Minimal levels of gas processing, and used for heating, process heating, boiler feed;
- *Medium-Grade LFG*: increased levels of processing to remove volatile organic compounds (VOC's), sulfur and moisture, and used in a wider variety of boilers and process heating applications; electricity generation with reciprocating engines, and simple and combined cycle gas turbines;
- *High-Grade LFG*: processing extends beyond that of medium-grade LFG and includes the separation of the gas into its two major components, methane (CH_4) and carbon dioxide (CO_2), and uses include: pipeline quality gas; electrical generation; sale of CO_2 ; methanol production; fuel cells; and vehicle fuel.

Although LFG technologies and utilization have been available for several years, it is only recently that modifications to the existing technologies and development of new technologies are beginning to make LFG utilization efficient enough to become economically viable. In fact, of the possible LFG utilization applications, five main ways to recover energy from Landfill Gas (LFG) are hereafter mentioned and suggested: direct heating, electricity generation, chemical feedstock, purification to pipeline quality gas, and heat recovery.

VI. LANDFILL GAS ELECTRICITY GENERATION

Landfill Gas (LFG) can be converted to electricity through a variety of means. The most common method currently employed involves the use of internal combustion reciprocating engines coupled with a generator. Other mature technologies are gas combustion turbine engines, simple cycle steam turbines, and combined cycle plants. New technologies include microturbines and fuel cells.

VII. SELECTION FACTORS FOR ELECTRICITY GENERATION

Several factors must be evaluated when considering generating electricity with LFG, whether the technology involves internal reciprocating engines, gas turbines, combined cycle, microturbines or steam turbines. Electricity conversion efficiency, which is an indication of what portion of the energy value of the LFG can be converted into electrical power, varies with each technology.

Other important factors that must be considered when deciding whether or not to utilize the LFG for electricity generation include availability, installation cost, operation and maintenance (O&M) costs, and emissions, all of which are site specific [10]. Beside these factors, capacity and configuration of landfills also affect the viability of LFG-to-Electricity projects, because they have an impact on the cost for materials, installation and maintenance of LFG collection system, and other factors.

After the annual gas generation is calculated, the gross power generation potential (GPGP) in a given year (in KW) can be calculated [4].

$$GPGP_T = \frac{LFG_T \eta_{col} E_c}{(365)(24)H_r} \quad (3)$$

where η_{col} is the collection system efficiency, typically 85%; E_c is the energy content of landfill gas, typically 500 Btu/m³; and H_r is the heat rate of equipment.

The net power generation potential (NPGP) is then estimated by subtracting the parasitic loads. This is energy lost by auxiliary equipment and is usually 2% of the gross power generation potential for IC Engines and 6% for gas or steam turbines, so that NPGP is 94-98% of GPGP. The annual electricity generated (in KWh) is then estimated [4].

$$AEG = NPGP (24)(365)90\% \quad (4)$$

where 24 is hours per day, 365 is days per year, and 90% represents the assumed average percentage of the time in a year that the equipment is producing electricity at its rated capacity (net of maintenance, downtime, etc).

VIII. ECONOMICS OF ELECTRICITY GENERATION

Based on the landfill gas (LFG) utilization and electricity generation technologies earlier mentioned and developed, the economics of electricity generation based on the US. EPA's recent data (all estimates are in 2010 dollars) [9], is summarized in the following table:

TABLE IV
LFG UTILIZATION TECHNOLOGIES AND TYPICAL FLOW AND POWER RANGES

Technology	Typical Flow Range [cfm]	Preferred Plant Size [MW]	Electrical Conversion efficiency %	Capital Costs [\$ /KW]	Annual O&M Costs [\$ /KW /Yr]
IC engines	300-1100	0.8-3	25-35	1,700-2,300	180-210
Gas turbines	1,300-2,100	1-10	20-28	1,400	130
Microturbines	20-200	0.03-0.25	25-30	5,500	380

IX. ECONOMIC EVALUATION OF LANDFILL GAS TO ENERGY (LFGTE) PROJECTS

Obviously, even though collecting and flaring LFG reduces the amount of methane (CH₄) in emissions, but converting the methane into energy conserves fossil fuels and brings about economic benefits. In fact, the economic viability of LFGTE projects depends on a number of factors, such as LFG quality, local energy prices, and choice of equipment. Previous researchers have undertaken studies on the economic assessment of LFGTE projects considering either the Internal Recovery Rate (IRR) or the Pay Back Time (PBT, which is defined as the amount of time needed for paying back the investment) as useful economic parameters or indicators.

For addressing this issue, many researchers have developed different economic models to evaluate the economic assessment of generating electricity from LFG for utility sale. These models mostly center round the capital and annual operation and maintenance costs. The economics of implementing a landfill gas to energy recovery and utilization projects is also addressed and evaluated using the same basis of concepts. But in this work, the only one option considered when performing this study, is the development of microturbines power plant taking into account the fact that they offer clean, quiet operation as compared to traditional IC engines, and particularly the ability to add and remove microturbines as gas flow increases or decreases.

The economic assessment of a landfill gas to energy (LFGTE) projects presented here is performed using Hybrid Optimization Model of Electric Renewable (HOMER) software. Four different case studies are possible viz., a simple stand-alone system (electricity only), a stand-alone combined heat and power (CHP) system, a simple grid connected system (electricity only) and a grid-connected combined heat and power (CHP) system. For this work, only one case-study or scenario is performed. In this assessment, considering options available in HOMER, certain costs such as costs of planning construction, LFG collection, flaring etc., have not been taken into account. The costs optimization for the LFGTE project aims at determining the best possible system configuration depending on the size of equipment, number of components and primary load to be served. Optimization criteria is based on the Net Present Cost (NPC) since it represents the total cost of the project including all cost components such as operation and maintenance (O&M),

replacement, fuel, total capital costs, etc. In addition, the economic evaluation performed in this work is assessed without carbon dioxide (CO₂) credits allocated.

The showcase configuration system (Stand-Alone LFGTE System) developed for the purpose of this study comprises the following:

1. *Four sets of microturbines*, manufactured either by Capstone Turbine Corporation or either G-Team, suitable to match the installed capacity determined from the Bellville South methane gas and LFGTE system configuration of Fig.2, and to match the on-site load demand and the primary load which is basically the Cape Town load profile represented in figure 3. In the size to consider, the ranges of CR1000 and 3000 KW (from G-Team) are considered in respect to the economics of electricity generation summarized in table V.
2. *Primary load1* which, in this case study is composed of the on-site load demand (load that have to be served at the generation facility or on the site) and the City of Cape Town load profile.
3. *Converter*, functioning as both rectifier (converting AC to DC) and inverter (DC to AC). The size and the cost of either installing or replacing a converter including all associated costs are given the table VI.
4. *Battery (or Battery Bank)*, manufactured by First National Battery and of type CCA SAE 700, serves as a backup of power on the site in the case of blackout or system failure. The Battery Bank consists of one string composed of 10 Batteries of 12 volts each, and 102 Ah. The capital cost value and all other relevant details are also listed in Table VI.

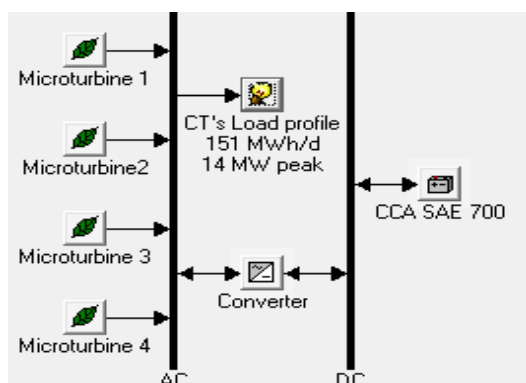


Fig.2. Modeling of stand-alone landfill gas to energy (LFGTE) system

The system component input parameters (number, size, and costs) are presented in the Table below:

TABLE VI
SYSTEM COMPONENTS SIZES AND COSTS

Component	Number	Size [KW]	capital [\$]	Replacement [\$]	O&M [\$ /hr]
Microturbine	4	1,000	5,500,000	4,500,000	380,000
		3,000	16,500,000	13,500,000	1,140,000
Converter	1	2,000	50,000	45,000	3,200
		5,000	125,000	112,500	8,000
Battery CCA SAE 700	1 string	120V	1,200/Bat.	800/Bat.	50/Bat.

X. SIMULATION RESULTS USING HOMER

The feasible system configurations sorted in descending order from the least to the highest NPC are presented in the Table A-I in the Appendix. The optimization criteria is based on the Net Present Cost (NPC) since it represents the total cost of the project including all cost components such as O&M, replacement, fuel, total capital costs, etc. This cost (NPC), and the same as the annualized cost can be sorted either by component cost, or by cost type. But in this work, only the NPC, Cost of Energy (COE), and emissions will be given and looked at.

Fig.3. Hourly data of microturbine 1 output power within a month

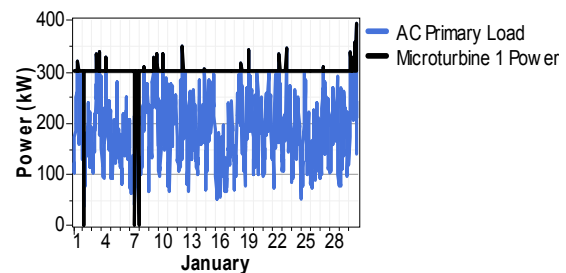


TABLE V
SYSTEM COMPONENTS SIZES AND COSTS

Pollutants	Emissions [Kg/yr]
Carbon dioxide (CO ₂)	11,796
Carbon monoxide (CO)	46,8
Unburned hydrocarbons	5,18
Particulate matter	3,53
Sulfur dioxide (SO ₂)	0
Nitrogen oxide (NO _x)	418

XI. ENVIRONMENTAL IMPACTS OF LANDFILL GAS TO ENERGY PROJECTS DEVELOPMENT

The generation of electricity, and the consumption of energy in general, result in consequences to the environment. But still environmental concerns have generally received a low priority when planning for an energy project. Nowadays, environmental concerns are increasingly becoming an important factor in the making of decisions on energy [11]. Thus, energy demand must be managed to cope not only with energy, but with environmental problems, as well.

Like the environmental impacts of conventional energy; the generation, extraction and use of energy through LFGTE projects is also generally and mostly concerned with the negative effects on the environment. Surely, these negative effects or environmental problems linked to waste disposal in landfills are related to water pollution, unpleasant odors, fire and explosion, asphyxiation, vegetation damage, and greenhouse gases (GHGs) emissions [8], [12-13]. However, using landfill gas (LFG) as another source of energy, particularly introducing an LFGTE project, decreases the environmental impacts from solid waste Landfilling through enhanced degradation process, and improved control and handling of landfill emissions. In fact, this results generally in the following environmental benefits: reducing local air

pollution, odor, risk of fire or explosion, greenhouse gases emissions, and impacts on local vegetation.

XII. CONCLUSION

Possibility of health and environmental hazards caused by growing amounts of municipal solid waste (MSW) are increasing with rapid population and economic growth and change in living standards. This has not only called for its proper management, but has also paved a way to be utilized as a low-carbon source for localized small to medium scale energy generation in LFG to energy systems.

In order to clearly determine what is the best option and cost-effective system configuration among those presented in table VI; numerous assumptions can be made. Include in these assumptions are: the total Net Present Cost (NPC), Cost of system operation and maintenance (O&M), Cost of Energy(COE), and unit sale price, duration and interest of capital financing, Emissions, and some others,...

The simulation results of a Stand-Alone LFGTE system indicate that this case presents a quite reasonable NPC, lower operating and maintenance (O&M) cost, but higher Cost of Energy (COE). And yet, this system still emits or releases carbon dioxide (CO₂) and any other pollutants such as CO, SO₂ and NO_x at reasonable and acceptable proportion.

As it can absolutely be seen from all possible system configurations simulated, only the first one appears to be the cost-optimized and best of the best options when considering Initial Capital Cost (ICC), O&M cost, NPC, and COE.

Finally, given the fact that converting the methane gas into a renewable energy source conserves fossil fuel and makes good economical sense, this study sets out to determine whether the development of an LFGTE project based on the Bellville South landfill site and using microturbine technology is technically and economically feasible. The results of the assessment carried out in this work have shown that the Bellville South landfill site is qualified and proven to be technically viable for being transformed into an LFGTE project.

XIII. REFERENCES

- [1] V.N. Popov, H. Power, Landfill emission of gases into the atmosphere: boundary element analysis. WIT press; 1999.
- [2] M.M. Schumacher, Landfill Methane Recovery. Noyes data corporation, Park Ridge, NJ. 1983.
- [3] P. Jaramillo, H.S. Matthew, Landfill-Gas-To-Energy Projects: Analysis of Net Private and Social. Environment, Science & Technology, 25 July, 2005
- [4] US Environmental Protection Agency, Government Printing Office, Turning a Liability into an Asset; Landfill Methane Outreach Program, Washington, DC, 1996.
- [5] D. Augenstein and J. Pacey, "Landfill Gas Energy Utilization: Technology options and case studies", prepared by EMCON Associates, San Jose, CA. Prepared for US. Environmental Protection Agency, Office of Research and Development. EPA-600/R-92-116. Washington, DC, June 1992.
- [6] W. Rattanawiboonson: Improvement of Predictable Model of Methane recovery within Landfill, Thesis, 2005.
- [7] Waste Characterization Surveys in the City of Town, 2nd Draft. Western Province Department of Environment and Development Planning, 2007.
- [8] V.N. Popov, A new landfill system for cheap landfill gas purification, Renewable Energy 30, 1025-1029, 2005.

- [9] US. Environmental Protection Agency. "Project Technology Options." LFG Energy Project Development Handbook, 9 September 2009.
- [10] R. Conestoga & Associates: Handbook for the preparation of Landfill Gas to Energy Projects in Latin America and the Caribbean. World Bank ESMAP, N° 019399 (6), January 2004.
- [11] Department of the Environment UK. Landfill gas-waste management paper n°27. London: Crown Copyright; 1995.
- [12] Department of the Environment UK. Landfill design, construction and operational practice- waste management paper n°26B. London: Crown Copyright; 1995.
- [13] US. Environmental Protection Agency, 2006 update: The state of US Landfill Gas Utilization Projects by Rachel Goldstein, Washington, DC, pp.13, 2006.
- [14] <http://www.math.about.com/od/formulas/a/compound.htm>. Compound interest formula.

XIV. APPENDIX

TABLE A-I

SYSTEM CONFIGURATIONS SIMULATIONS SORTED BY LIST OF COST-EFFECTIVENESS OF FEASIBLE SYSTEM DESIGNS

MT ₁ [KWh]	MT ₁ [KWh]	MT ₁ [KWh]	MT ₁ [KWh]	MT ₁ [KWh]	Biomass [t]	Renewable Fractions	COE [\$/KWh]	Total NPC [\$]	Operating Cost [\$/yr]	Initial Capital[\$]	Efficiency Measures	Dispatched Strategy	CONVER [KWh]	CCA SAE 700	MT ₁ [KWh]	MT ₁ [KWh]	MT ₁ [KWh]	MT ₁ [KWh]
1,000	-	-	-	-	7,199	1.00	3,863	71,711,560	2,945,550	38,551,200	Yes	CC	2,000	10	-	-	-	-
-	-	-	-	-	7,199	1.00	3,863	71,711,560	2,945,550	38,551,200	Yes	CC	2,000	10	-	-	-	-
-	-	-	-	-	7,199	1.00	3,863	71,711,560	2,945,550	38,551,200	Yes	CC	2,000	10	-	-	-	-
-	-	-	-	-	7,199	1.00	3,863	71,711,560	2,945,550	38,551,200	Yes	CC	2,000	10	-	-	-	-
-	-	-	-	-	7,199	1.00	4,135	76,777,784	2,907,108	44,051,200	Yes	CC	2,000	10	-	-	-	-
-	-	-	-	-	7,199	1.00	4,135	76,777,784	2,907,108	44,051,200	Yes	CC	2,000	10	-	-	-	-
-	-	-	-	-	7,199	1.00	4,408	81,844,000	2,868,486	49,551,200	Yes	CC	2,000	10	-	-	-	-
-	-	-	-	-	7,199	1.00	4,681	86,910,216	2,829,954	55,051,200	Yes	CC	2,000	10	-	-	-	-
-	-	-	-	-	7,199	1.00	4,689	87,060,528	2,836,644	55,126,200	Yes	CC	2,000	10	-	-	-	-
-	-	-	-	-	7,199	1.00	4,689	87,060,528	2,836,644	55,126,200	Yes	LF	5,000	10	-	-	-	-
-	-	-	-	-	7,199	1.00	4,689	87,060,528	2,836,644	55,126,200	Yes	CC	5,000	10	-	-	-	-
-	-	-	-	-	7,199	1.00	4,689	87,060,528	2,836,644	55,126,200	Yes	LF	5,000	10	-	-	-	-
-	-	-	-	-	7,199	1.00	4,954	91,976,432	2,791,422	60,551,200	Yes	CC	2,000	10	-	-	-	-
-	-	-	-	-	7,199	1.00	5,227	97,042,648	2,752,590	66,051,200	Yes	CC	2,000	10	-	-	-	-
-	-	-	-	-	7,199	1.00	5,227	97,042,648	2,752,590	66,051,200	Yes	LF	2,000	10	-	-	-	-
-	-	-	-	-	7,199	1.00	5,500	102,108,86	2,714,459	71,551,200	Yes	CC	2,000	10	-	-	-	-

The row in yellow represents the optimized results of the sensitivity analysis. Dispatch strategy: the dispatch strategy determines how the system charges the Battery Bank.

➤ Load following: LF; Cycle charging: CC